

HW Assignment 1

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Question 1a

h1. $\forall P(\exists T(R(P, T)))$

where:

- P = project
- T = research topic
- $R(X, Y)$ = project X is related to research topic Y

h2. $\forall P\forall T\forall S((R(P, T) \wedge W(P, S)) \implies L(S, T))$

where:

- P = project
- T = research topic
- S = student
- $W(A, B)$ = student B works on project A
- $L(C, D)$ = student C studies research topic D

c3. $\forall P(V(A, P) \implies \exists S(W(P, S)))$

where:

- P = project
- A = academic supervisor
- S = student
- $V(X, Y)$ = supervisor X supervises project Y

Given facts:

- h4. $W(\text{proj1}, \text{Cris})$
- h5. $R(\text{proj2}, \text{AI}) \wedge R(\text{proj2}, \text{Logic})$
- h6. $V(\text{Alice}, \text{proj2})$

Question 1b

1. $\forall P(\exists T(R(P, T)))$

skolemisation: $\forall P(R(P, f(P)))$

generalisation: $R(P, f(P))$

2. $\forall P\forall T\forall S((R(P, T) \wedge W(P, S)) \implies L(S, T))$

eliminate implications: $\forall P\forall T\forall S(\neg(R(P, T) \wedge W(P, S)) \vee L(S, T))$

push negation: $\forall P\forall T\forall S((\neg R(P, T) \vee \neg W(P, S)) \vee L(S, T))$

generalisation: $(\neg R(P, T) \vee \neg W(P, S)) \vee L(S, T)$

remove brackets: $\neg R(P, T) \vee \neg W(P, S) \vee L(S, T)$

3. $\forall P(V(A, P) \implies \exists S(W(P, S)))$

eliminate implications: $\forall P(\neg V(A, P) \vee \exists S(W(P, S)))$

skolemisation: $\forall P(\neg V(A, P) \vee W(P, f(P)))$

generalisation: $\neg V(A, P) \vee W(P, f(P))$

Final clauses:

- c1: $W(\text{proj1}, \text{Cris})$
- c2: $R(\text{proj2}, \text{AI})$
- c3: $V(\text{Alice}, \text{proj2})$
- c4: $R(P, f(P))$
- c5: $\neg R(P, T) \vee \neg W(P, S) \vee L(S, T)$
- c6: $\neg V(A, P) \vee W(P, f(P))$
- c7: $R(\text{proj2}, \text{Logic})$

Question 2a

Prove: $(A \vee B) \wedge \neg B \implies A$

let:

$h1 = A \vee B$

$h2 = \neg B$

$$\frac{\frac{\frac{A \vee B}{(h1)} \quad [A]^1}{A} \quad \frac{\frac{[B]^2 \quad \overline{\neg B} (h2)}{\perp} (\neg E)}{A} (\vee E)^{1,2}}{(A \vee B) \wedge \neg B \implies A} (\implies I)^{h1, h2}$$

Question 2b

Prove: There is a research topic that Cris studies

Prove (FOL form): $\exists T(L(\text{Cris}, T))$

$$\frac{\frac{\forall P(\exists T(R(P, T)))}{\exists T(R(\text{proj1}, T))} (\forall E) \quad \frac{\frac{\frac{[R(\text{proj1}, t_1)]^1 \quad \overline{W(\text{proj1}, \text{Cris})}}{R(\text{proj1}, t_1) \wedge W(\text{proj1}, \text{Cris})} (\wedge I) \quad \frac{(R(\text{proj1}, t_1) \wedge W(\text{proj1}, \text{Cris}))}{\implies L(\text{Cris}, t_1)} (\implies E)}{L(\text{Cris}, t_1)} (\exists I)}{\exists T(L(\text{Cris}, T))} (\exists E^1)}{\exists T(L(\text{Cris}, T))} (\exists E)$$

Question 2c

Prove: There is a student studying AI

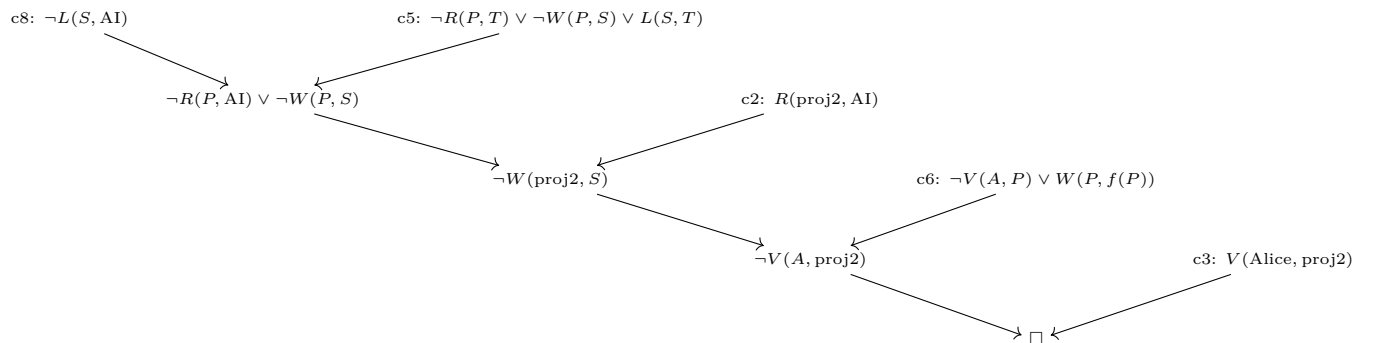
Prove (FOL form): $\exists S(L(S, \text{AI}))$

negation: $\neg \exists S(L(S, \text{AI}))$

move negation inwards: $\forall S(\neg L(S, \text{AI}))$

generalisation: $\neg L(S, \text{AI})$

c8: $\neg L(S, \text{AI})$



Question 3 (Text Version)

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1    connected(perth, bull_creek, perth_mandurah).
2    connected(bull_creek, cockburn, perth_mandurah).
3    connected(elizabeth_quay, canning_bridge, perth_mandurah).
4    connected(murdoch, aubin_grove, perth_mandurah).
5    connected(bull_creek, murdoch, perth_mandurah).
6    connected(perth, burswood, perth_armadale).
7    connected(cannington, beckenham, perth_armadale).
8    connected(kelmscott, byford, perth_armadale).
9    connected(victoria_park, cannington, perth_armadale).
10   connected(armadale, byford, perth_armadale).
11   connected(burswood, victoria_park, perth_armadale).
12   connected(beckenham, kelmscott, perth_armadale).
13   connected(kelmscott, armadale, perth_armadale).
14
15   closed(murdoch).
16
17
18   station_on_line(S, L) :-
19       connected(S, _, L);
20       connected(_, S, L).
21
22   before_on_line(S, T, L) :-
23       connected(S, T, L).
24   before_on_line(S, T, L) :-
25       connected(S, X, L),
26       before_on_line(X, T, L).
27
28   terminal(S, L) :-
29       connected(S, _, L),
30       \+ connected(_, S, L).
31   terminal(S, L) :-
32       connected(_, S, L),
33       \+ connected(S, _, L).
34
35   route_on_line(S, T, L, [S, T]) :-
36       connected(S, T, L),
37       \+ closed(S),
38       \+ closed(T).
39
40   route_on_line(S, T, L, [S | Rest]) :-
41       connected(S, X, L),
42       \+ closed(S),
43       route_on_line(X, T, L, Rest).
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